

Model Development Phase Template

Date	20 June 2025
Team ID	SWTID1749707061
Project Title	Online Payment Fraud Detection Using Machine Learning
Maximum Marks	10 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training code will be showcased in the future through a screenshot. The model validation and evaluation report will include a summary and training and validation performance metrics for multiple models, presented through respective screenshots.

Initial Model Training Code (5 marks):

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,random_state=0,test_size=0.2)
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder

# 1. Drop identifiers that can't be converted to float
df_cleaned = df.drop(columns=['nameOrig', 'nameDest'], errors='ignore')

# 2. Encode categorical features
if 'type' in df_cleaned.columns:
    le = LabelEncoder()
    df_cleaned['type'] = le.fit_transform(df_cleaned['type'])

# 3. Replace inf values and drop/handle NaNs
df_cleaned.replace([np.inf, -np.inf], np.nan, inplace=True)
df_cleaned.dropna(inplace=True)

# 4. Split into features and target
X = df_cleaned.drop(columns=['isFraud'])
y = df_cleaned['isFraud']

# 5. Train-test split
x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 6. Train model
rfc = RandomForestClassifier()
rfc.fit(x_train, y_train)

# 7. Predict & evaluate
y_test_predict1 = rfc.predict(x_test)
test_accuracy = accuracy_score(y_test, y_test_predict1)

print("Test Accuracy:", test_accuracy)
```

Test Accuracy: 0.9997288846313734

```
[ ] print(classification_report(y_test,y_test_predict1))
```

#Decision Tree

```
from sklearn.tree import DecisionTreeClassifier
dtc=DecisionTreeClassifier()
dtc.fit(x_train,y_train)

y_test_predict2=dtc.predict(x_test)
test_accuracy=accuracy_score(y_test,y_test_predict2)
test_accuracy
```

#Extra Trees

```
from sklearn.ensemble import ExtraTreesClassifier
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.datasets import load_iris

data = load_iris()
X = data.data
y = data.target

x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

etc = ExtraTreesClassifier()
etc.fit(x_train, y_train)

y_test_predict3 = etc.predict(x_test)
test_accuracy = accuracy_score(y_test, y_test_predict3)
print(test_accuracy)
```

#SVC

```
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
svc=SVC()
svc.fit(x_train,y_train)
y_test_predict4=svc.predict(x_test)
test_accuracy=accuracy_score(y_test,y_test_predict4)
test_accuracy
```

```
#xgb
import xgboost as xgb
xgb1 = xgb.XGBClassifier()
xgb1.fit(x_train,y_train1)

y_test_predict5=xgb1.predict(x_test)
test_accuracy=accuracy_score(y_test1,y_test_predict5)
test_accuracy

def compareModel():
    print("train accuracy for rfc", accuracy_score(y_train_predict1,y_train))
    print("test accuracy for rfc", accuracy_score(y_test_predict1,y_test))
    print("train accuracy for dtc", accuracy_score(y_train_predict2,y_train))
    print("test accuracy for dtc", accuracy_score(y_test_predict2,y_test))
    print("train accuracy for etc", accuracy_score(y_train_predict3,y_train))
    print("test accuracy for etc", accuracy_score(y_test_predict3,y_test))
    print("train accuracy for svc", accuracy_score(y_train_predict4,y_train))
    print("test accuracy for svcc", accuracy_score(y_test_predict4,y_test))
    print("train accuracy for xgb1", accuracy_score(y_train_predict5,y_train1))
    print("test accuracy for xgbi", accuracy_score(y_test_predict5,y_test1))
```

Model Validation and Evaluation Report (5 marks):

Model	Classification Report	F1 Score	Confusion Matrix																																										
Random Forest	<table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>is Fraud</td><td>0.98</td><td>0.80</td><td>0.88</td><td>1587</td></tr><tr><td>is not Fraud</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1270934</td></tr><tr><td>accuracy</td><td></td><td></td><td>1.00</td><td>1272521</td></tr><tr><td>macro avg</td><td>0.99</td><td>0.90</td><td>0.94</td><td>1272521</td></tr><tr><td>weighted avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1272521</td></tr></tbody></table>		precision	recall	f1-score	support	is Fraud	0.98	0.80	0.88	1587	is not Fraud	1.00	1.00	1.00	1270934	accuracy			1.00	1272521	macro avg	0.99	0.90	0.94	1272521	weighted avg	1.00	1.00	1.00	1272521	98.4%	<table><thead><tr><th>col_0</th><th>is Fraud</th><th>is not Fraud</th></tr></thead><tbody><tr><td>isFraud</td><td></td><td></td></tr><tr><td>is Fraud</td><td>1271</td><td>316</td></tr><tr><td>is not Fraud</td><td>29</td><td>1270905</td></tr></tbody></table>	col_0	is Fraud	is not Fraud	isFraud			is Fraud	1271	316	is not Fraud	29	1270905
		precision	recall	f1-score	support																																								
	is Fraud	0.98	0.80	0.88	1587																																								
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is Fraud	1271	316																																											
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Decision tree	precision	recall	f1-score	support	98.6%	col_0 is Fraud is not Fraud 	
	is Fraud	0.90	0.89	0.89		1587	isFraud 
	is not Fraud	1.00	1.00	1.00		1270934	
	accuracy			1.00		1272521	
	macro avg	0.95	0.94	0.95		1272521	
	weighted avg	1.00	1.00	1.00		1272521	
Extra Tree	precision	recall	f1-score	support	99.01%	col_0 is Fraud is not Fraud	
	is Fraud	0.99	0.79	0.88		1587	isFraud
	is not Fraud	1.00	1.00	1.00		1270934	
	accuracy			1.00		1272521	
	macro avg	1.00	0.89	0.94		1272521	
	weighted avg	1.00	1.00	1.00		1272521	
Support Vector Machine	precision	recall	f1-score	support	99.06%	col_0 is Fraud is not Fraud	
	is Fraud	0.96	0.42	0.58		1587	is Fraud
	is not Fraud	0.99	1.00	0.99		1270934	is Fraud 669 918
	accuracy			0.99		1272521	is not Fraud 81 1270853
	macro avg	0.97	0.71	0.78		1272521	
	weighted avg	0.99	0.99	0.99		1272521	
XG Boost	precision	recall	f1-score	support	99.97%	Col_0 is Fraud is not Fraud	
	is Fraud	0.87	0.88	0.88		1587	is Fraud
	is not Fraud	1.00	1.00	1.00		1270934	is Fraud 1398 189
	accuracy			1.00		1272521	is not Fraud 215 1270719
	macro avg	0.94	0.94	0.94		1272521	
	weighted avg	1.00	1.00	1.00		1272521	